IIT Kharagpur ML Security Hackathon - Problem 1
Clean-Only HPC Backdoor Anomaly Detection

Objective

Detect inference traces that differ from benign HPC behavior, with no backdoor traces available during training. The supplied file has 800 clean inferences and three hardware performance counters.

Data preparation

Features: cycles, cache-references, LLC-loads. The unnamed sequential column is a row index and is dropped. Features are log(cycles), log(cache-references/cycles), and log(LLC-loads/cache-references). Cache references and LLC loads have approximately 0.92 Pearson correlation in the supplied clean traces.

Proposed detector

A Gaussian mixture density is fitted to the joint transformed HPC vector. BIC selects the mixture size on the fitting split. The anomaly score is negative log likelihood. A separate clean calibration split sets the threshold at the configured 95th percentile. The artifact saves scaler, model, feature names, and threshold.

Evaluation design

Twenty-five deterministic clean-only splits: 60% fit, 20% calibration, 20% held-out clean evaluation. Baselines: (1) per-counter GMM proxy using the maximum standardized marginal negative log likelihood, and (2) joint GMM on log raw counter values. The proposed work-conditioned feature GMM is compared using identical splits.

Clean-only results: mean held-out FPR +/- standard deviation
Per-counter GMM proxy: 5.33% +/- 2.21%
Joint raw-log GMM: 4.60% +/- 1.78%
Work-conditioned feature GMM: 4.75% +/- 1.81%

Interpretation and limits

These results measure clean false alarms only. Small FPR differences do not demonstrate superior backdoor detection. No attack examples were provided, so TPR, attack FPR, F1, and AUROC cannot be calculated. Triggered proxy runs, if collected, must be reported as separate evaluation. Calibration assumes future clean traces are comparable to calibration traces; it does not guarantee the same FPR under distribution shift.

Related work

Alam, Wang, Maniatakos, DATE 2024, Detecting Backdoor Attacks in Black-Box Neural Networks through Hardware Performance Counters.
https://past.date-conference.com/proceedings-archive/2024/DATA/1142_pdf_upload.pdf

Reproduction commands

```
pip install -r requirements.txt
python src/train_detector.py --input data/clean_hpc.csv --output artifacts
python src/evaluate_clean.py --input data/clean_hpc.csv --output artifacts
```